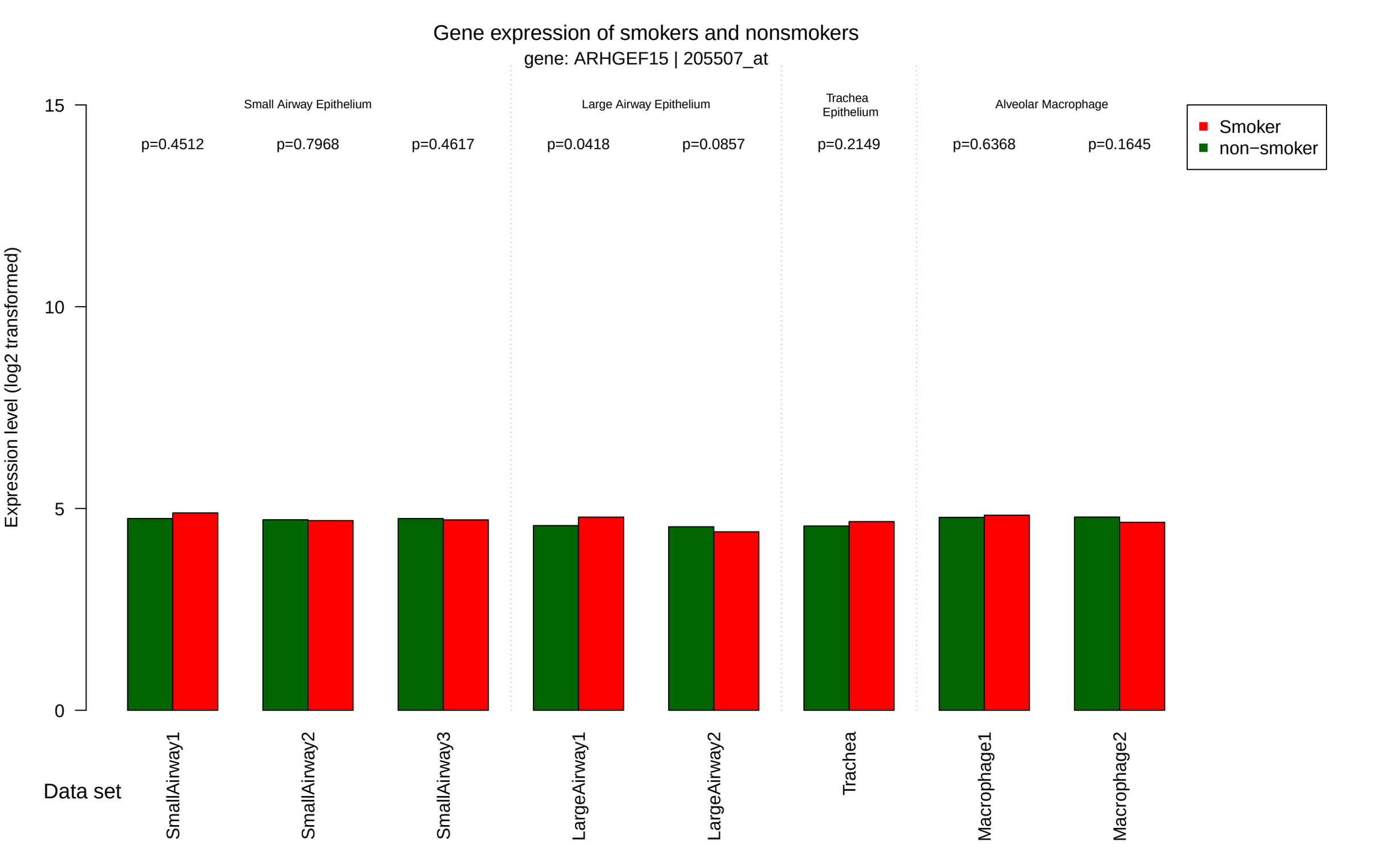
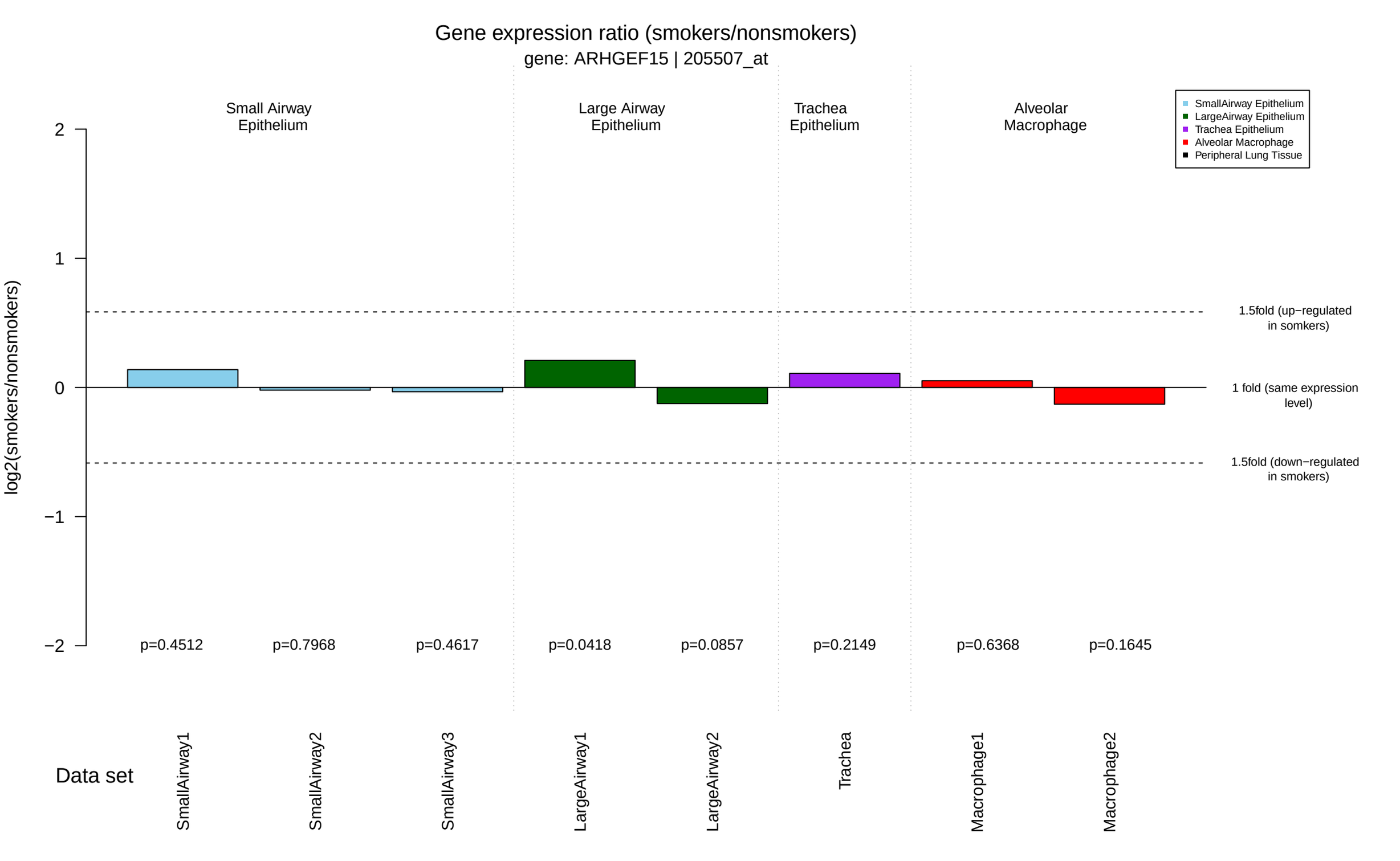
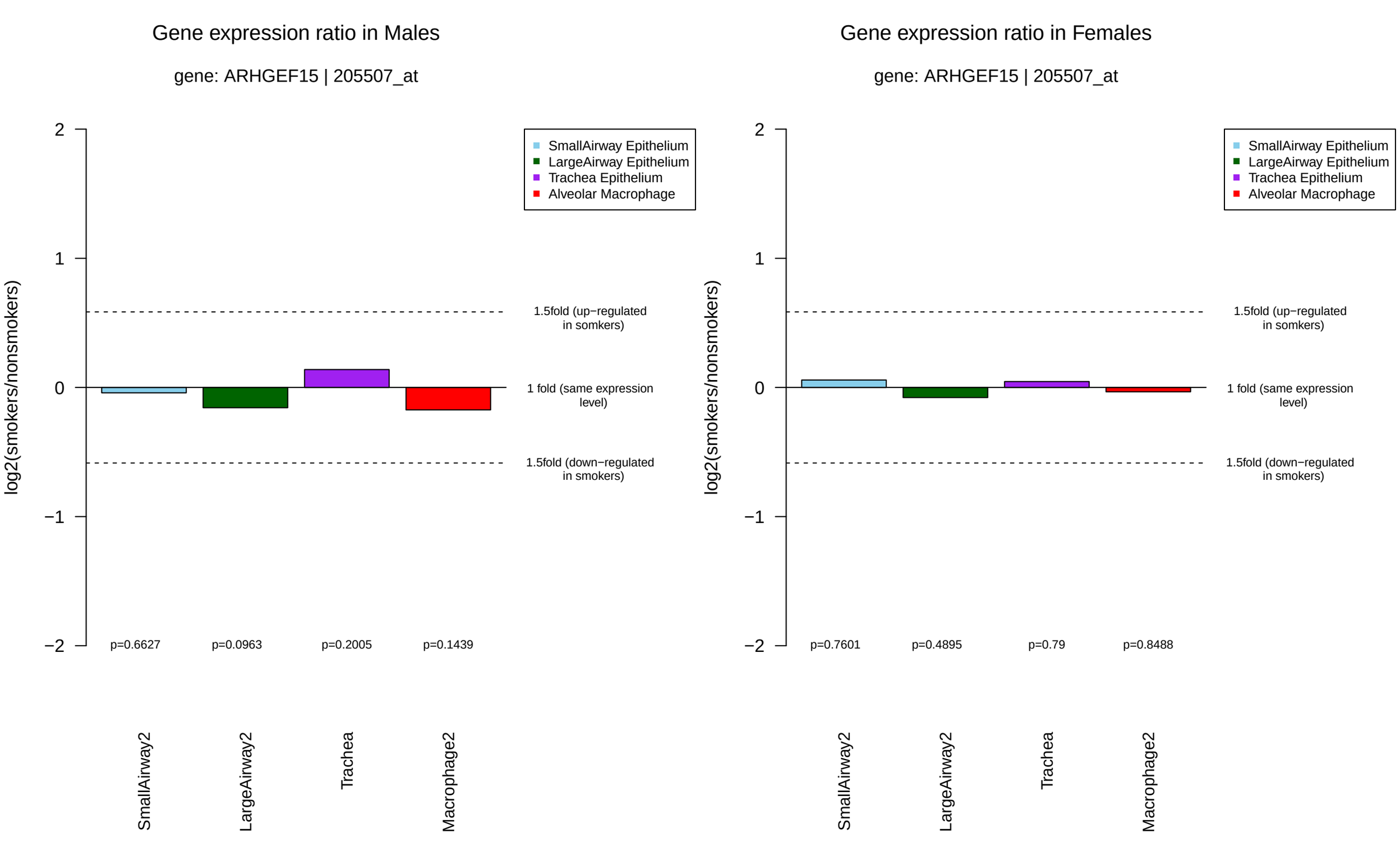
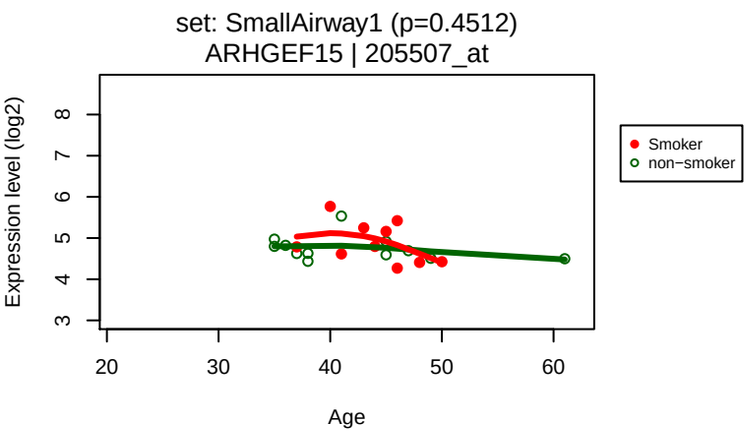
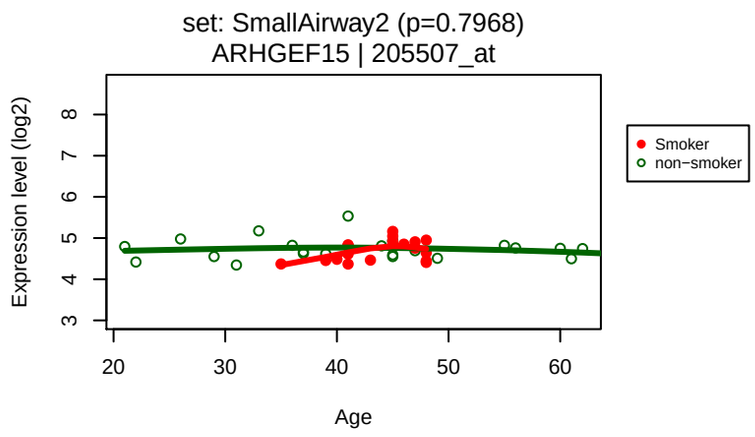
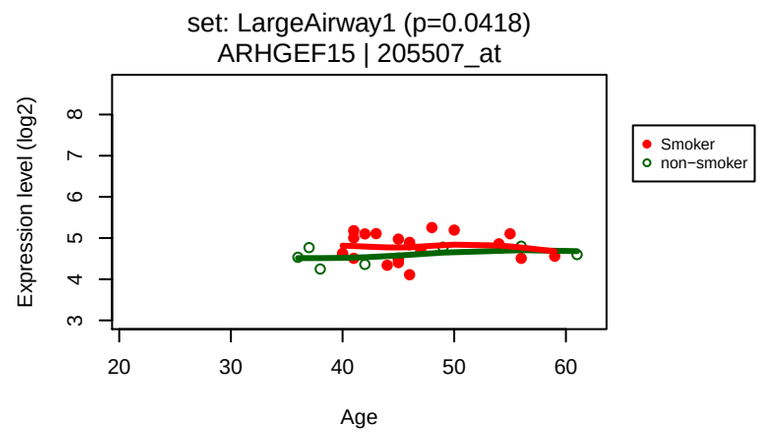

Gene expression of smokers and nonsmokers
gene: ARHGEF15 | 205507_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4512
p=0.7968
p=0.4617
p=0.0418
p=0.0857
p=0.2149
p=0.6368
p=0.1645
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

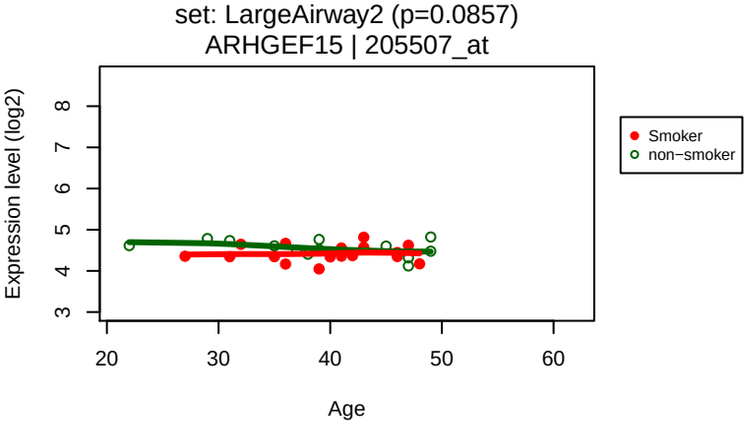

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF15 | 205507_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4512
p=0.7968
p=0.4617
p=0.0418
p=0.0857
p=0.2149
p=0.6368
p=0.1645
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

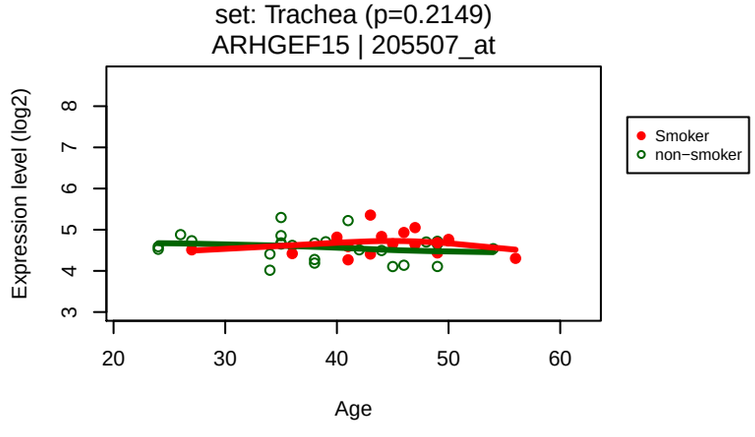

Gene expression ratio in Males
gene: ARHGEF15 | 205507_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6627
p=0.0963
p=0.2005
p=0.1439
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF15 | 205507_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7601
p=0.4895
p=0.79
p=0.8488
SmallAirway2
LargeAirway2
Trachea
Macrophage2

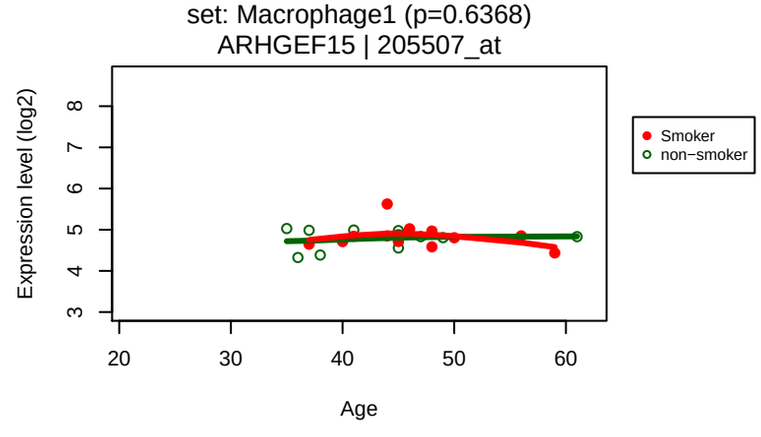

set: SmallAirway1 (p=0.4512)
ARHGEF15 | 205507_at
Expression level (log2)
Age
Smoker
non−smoker

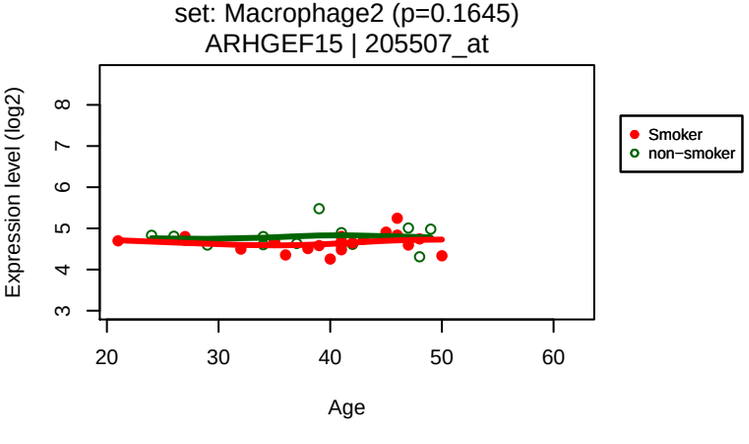

set: SmallAirway2 (p=0.7968)
ARHGEF15 | 205507_at
Expression level (log2)
Age
Smoker
non−smoker

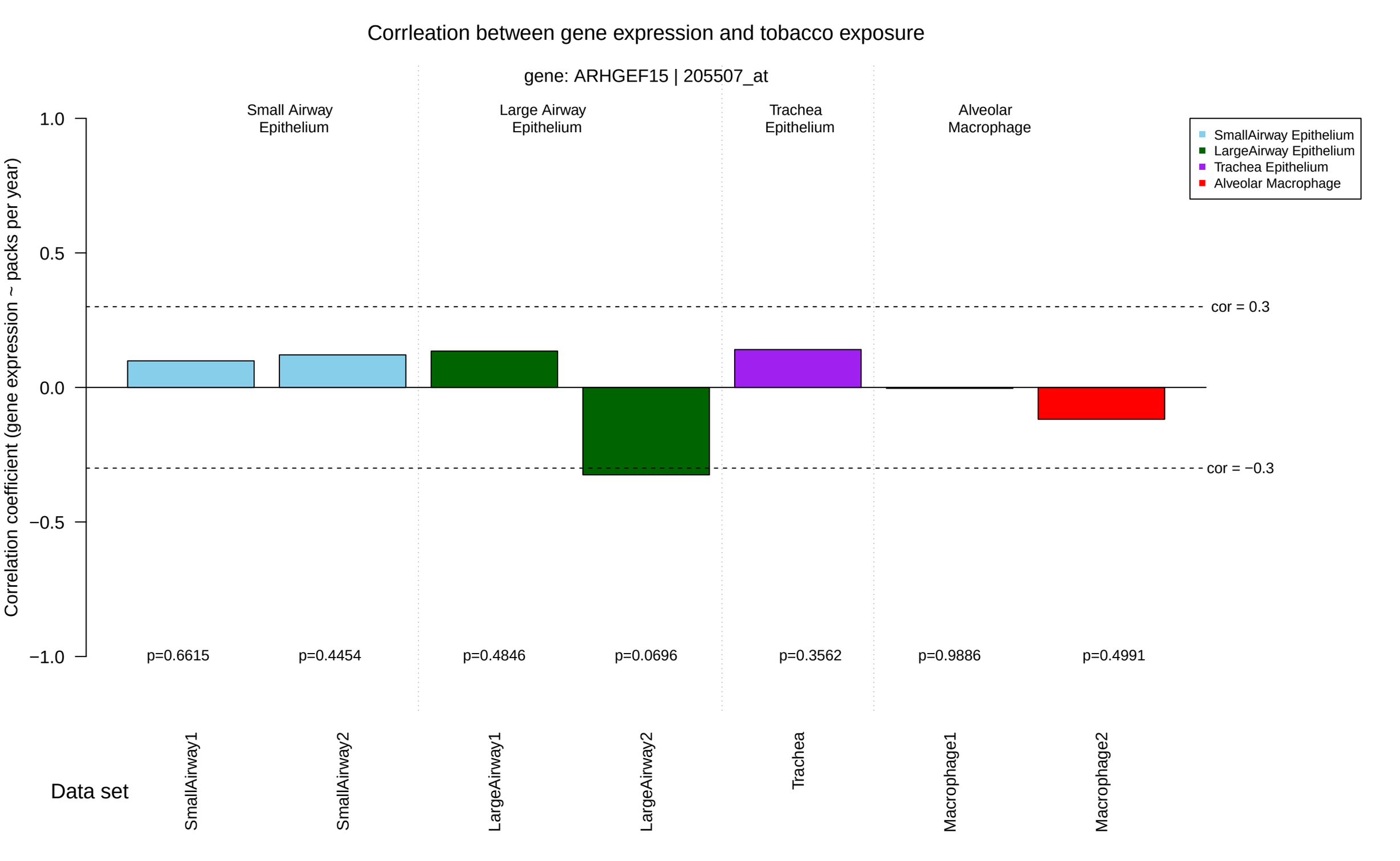

set: LargeAirway1 (p=0.0418)
ARHGEF15 | 205507_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0857)
ARHGEF15 | 205507_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2149)
ARHGEF15 | 205507_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6368)
ARHGEF15 | 205507_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1645)
ARHGEF15 | 205507_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF15 | 205507_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6615
p=0.4454
p=0.4846
p=0.0696
p=0.3562
p=0.9886
p=0.4991
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2